1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "wolf" even through no wolf is in sight. The villagers run to protect the flock, but the get really mad when thy realize the boy was playing a joke on them one night, the shepherd boy sees a real wolf approaching the flock and calls out, "wolf" The villagers refuse to be fooled

Act \ pre	wolf=yes	wolf=NO
wolf =y	TP wolf came warning	FN wolf not came
wolf =N	FP wolf not came warning	TN wolf not came warning came

2. Assume there are 100 Images 30 of them depict a cat the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat Image. It also predicts absence of a cat in 50 of the 70 no cat Images Identify TP, FP, FN, TN and calculate the precision and Recall and F1 score.

Actual \ predicted	cat=30	70 No cat
cat	TP=25	FN=5
No cat	FP=20	TN=50

$$\text{Accuracy} = \frac{(TP+TN)}{All}$$

$$\text{precesion} = \frac{TP}{TP+FP} = \frac{25}{25+20} = 0.55 = 55.56\%$$

$$\text{Recell} = \frac{TP}{TP+FN} = \frac{25}{25+5} = \frac{25}{30} = 0.83$$

$$F = \frac{2 \times 0.55 \times 0.83}{0.55 + 0.82} = \frac{0.913}{1.37} = 0.66 =$$

(2) An Email serive provider wants to classify emails as Spam or Not spam using features

- contains "offer" (yes/NO)
- contains "Free" (yes/NO)
- Email length (short/Long)

Sample Dataset:- Offer free length class, yes yes short spam
Yes No Long Not spam, NO yes short spam, NO, No Long Not Spam. Yes. Yes Long Spam, NO NO short Not Spam

- Total = 6

Spam = 3, Not Spam = 3

probability of spam = $P(spam) = \frac{3}{6} = \frac{1}{2}$

$p(not\ spam) = \frac{3}{6} = \frac{1}{2}$

conditional probability for each feature

Feature value	P(Feature/Spam)	P(Feature/Notspam)
offer = yes	$2/3 \approx 0.67$	$1/3 \approx 0.33$
Offer = NO	$1/3 \approx 0.33$	$2/3 \approx 0.67$
Free = yes	$3/3 = 1.00$	$0/3 = 0.00$
Free = NO	$0/3 = 0.00$	$3/3 = 1.00$
Length = Short	$2/3 = 0.67$	$1/3 = 0.33$
Length = long	$1/3 \approx 0.33$	$2/3 \approx 0.67$

Confusion Matrix problem:-

Actual class / predicted class	cancer = yes	cancer = no	Total
cancer = yes	90	210	300 P
cancer = no	140	9560	9700
Total	230	9770	10000 All

$$\text{Accuracy} = \frac{(TP + TN)}{ALL} = \frac{90 + 9560}{10000} = 96.5$$

$$\text{Error} = 1 - 96.5 = 0.035 = 3.5\%.$$

$$\text{Sensitivity} = \frac{TP}{P} = \frac{90}{300} = 0.3 = 30\%$$

$$\text{specificify} : \frac{TN}{N} = \frac{9560}{9700} = 98.5 = 98\%$$

$$\text{Recall precision} = \frac{TP}{TP + FP} = \frac{90}{90+210} = 0.3 = 30\%$$

$$\text{precision Recall} = \frac{TP}{TP+FN} : \frac{90}{90+140} = 39\%.$$

$$F = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

$$= \frac{2 \times 0.30 \times 0.39}{0.3 + 0.39} = \frac{0.234}{0.69} = 0.3391$$